

Fig. 5.34

PD between conductor A and neutral 'infinite' plane,

$$V_{AN} = \int_r^\infty \frac{q}{2\pi \epsilon_0} \cdot \frac{d}{dx} + \int_d^\infty \frac{-q}{2\pi \epsilon_0} \frac{dx}{x}$$

$$= \frac{q}{2\pi \epsilon_0} \left[\log_e \infty + \log_e \frac{1}{r} - \log_e \infty + \log_e d \right]$$

$$= \frac{q}{2\pi \epsilon_0} \log_e \frac{d}{r}$$

Similarly *pd* between conductor B and neutral 'infinite' plane,

$$V_{BN} = \int_r^\infty \frac{-q}{2\pi \epsilon_0} \frac{dx}{x} + \int_d^\infty \frac{q}{2\pi \epsilon_0} \frac{dx}{x} = \frac{-q}{2\pi \epsilon_0} \log_e \frac{d}{r} \text{ volts}$$

PD between conductors A and B,

$$V_{AB} = V_{AN} - V_{BN}$$

$$= \frac{q}{2\pi \epsilon_0} \log_e \frac{d}{r} - \frac{-q}{2\pi \epsilon_0} \log_e \frac{d}{r} = \frac{q}{\pi \epsilon_0} \log_e \frac{d}{r} \text{ volts}$$

Capacitance of the line,

$$C = \frac{q}{V_{AB}} = \frac{q}{\frac{q}{\pi \epsilon_0} \log_e \frac{d}{r}} = \frac{\pi \epsilon_0}{\log_e \frac{d}{r}} \text{ F/m} \qquad ...(5.40)$$